

MIGUEL ANTONIO BALIOG

Software Engineer

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SUMMARY

Software Engineer building fintech features for a global payment gateway: React and Vite front ends, Flask and PostgreSQL backends, and reconciliation tooling spanning Checkout financial actions, payouts, and a revenue calculator, backed by PostgreSQL schema migrations for transaction, invoice, and payment data. Also built out an internal Enterprise Dashboard and PTS analytics suite, including multi-agent search, payout archiving, and Merchant Terminal and Commissions views. Background further includes applied machine learning and computer vision research, including a two-stage YOLOv8 pipeline for Braille cell detection and recognition published in Proc. SPIE and benchmarked against CNN and ResNet architectures, plus earlier experience in data analysis and automation testing. Comfortable owning a feature end to end, from database schema through UI, independently or within a Scrum team.

WORK EXPERIENCE

Software Engineer II | Aqwire

April 2026 - Present

- Built a Checkout transaction-reconciliation system end to end (reports service, reconciliation controller/service, and a dedicated Checkout Payout page) to reconcile internal records against Checkout, a global payment gateway.
- Added financial-action and payout views with running totals, search, sorting, and cross-linked navigation across related payment, payout, and invoice pages.
- Built an initial revenue calculator and extended prepared settlements to handle PTS transactions.
- Authored PostgreSQL migrations for Checkout financial actions and payouts, including the invoice linkage used across reconciliation views.
- Streamlined an automated scraper for Maya transaction reports that retrieves the data and uploads it via SSH to a production database.

Software Engineer | Aqwire

March 2025 - April 2026

- Implemented payment archiving end to end (PostgreSQL functions, migration scripts, and Enterprise Dashboard UI), letting admins archive, unarchive, and filter payments.
- Built the Holidays feature (CRUD, PostgreSQL functions, filtering by year) and integrated it into System Settings and the dashboard overview.
- Delivered multi-agent search across the Enterprise Dashboard, Merchant Dashboard, and Enrollments, filtering transactions and enrollments by agent.
- Built out PTS Dashboard analytics for Admin, Agent, and Merchant views (overview panels, transaction charts, Merchant Terminals, Maya MID management, and Commissions pages) with a React/Vite front end, Python/Flask backend, and PostgreSQL for data management.
- Added invoice escrow tagging in PostgreSQL functions covering direct debit and autodebit card payments.

Data Analyst | Aclara Meters Philippines, Inc.

March 2024 - June 2024

- Assisted in the digitization of company data management tools.
- Digitized hard-copy records into Excel, building custom data-entry functionality with Visual Basic.
- Used Microsoft Excel to manage data, including developing a Material Resources Planning (MRP) system to support inventory management, data analysis, and logistics.

EDUCATION

Bachelor of Science in Computer Science, Major in Software Technology

De La Salle University - Manila | 2019 - 2024

SKILLS

Programming Languages: Python, Java, JavaScript, TypeScript, C, Fortran, Prolog, Visual Basic

Web Development: ReactJS, Angular, Vite, NodeJS, Flask, Tailwind CSS, CSS, HTML, HBS, SCSS

Database Management: PostgreSQL, MySQL, Redis, Firebase, MongoDB, Supabase

Cloud Platforms: AWS, Azure

Automation Testing: Selenium, JEST, Robot Framework

Machine Learning: YOLOv8, CNN, ResNet, Computer Vision, Object Detection

Data Modeling: Python, Jupyter, Pandas, Pydantic

Scripting: Bash, Shell

Mobile Development: Android Studio

Computer Literacy: Microsoft Office

PROJECTS

Transliterating Unified English and Filipino Braille to Romanized Text Using Image Processing

Published in Proc. SPIE (doi.org/10.1117/12.3055114). Transliterates photos of Grade 1 Braille modules into romanized text (Filipino and English) to help guardians of visually impaired students at the Philippine National School for the Blind.

- Designed a two-stage YOLOv8 pipeline for Braille cell detection and recognition (up to 99.2% top-1 / 100% top-5 accuracy on cell recognition), benchmarked against CNN and ResNet18/34/50.
- Built and annotated a custom image dataset using Roboflow and CVAT.

MediData - Medical Records App

- Angular and Supabase app for tracking patients, visits, and medications, with a Chart.js dashboard and offline support as an installable PWA.

CountIt - Personal Health and Nutrition Tracking Android Application

- Built using ESHA Nutrition API, Java, MySQL, and Firebase.

MIC (Music for Inspiring Creators) - Social Media Web Application

- Built with HTML, CSS, NodeJS, JavaScript, MongoDB, and Firebase.

CUISINA - E-Commerce Website for Kitchen and General Cooking Supplies

- Automation testing with Selenium (front-end), JEST and Robot Framework (back-end); manual testing with Microsoft Excel (front-end and back-end).
- Used Scrum Agile methodology and a CI/CD pipeline; managed team and task tracking through Linear.

LANGUAGES

English - Native

Filipino / Tagalog - Native